

I design and implement complex software systems: modular .NET platforms, backend services with payments and recovery, compilers, and program-analysis tools. My strongest areas are explicit contracts, state invariants, verifiable semantics, and carrying systems through to reproducible outcomes.

Platform architecture

UniversalToolchain/Wist2: independent packages, typed contracts, routes, manifests/locks, and lifecycle ownership

Production backend

Payments, subscriptions, roles, devices, PostgreSQL/SQLite, migrations, recovery, backup/restore, and rollback

Compiler / systems

Callable-first SSA, interpreter/CIL, LLVM, C++23, memory analysis, and x86-64 code generation

| EXPERIENCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                          | SKILLS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| 2024 — present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <div><div>UniversalToolchain / Wist2 — creator and primary developer</div><ul style="list-style-type: none"><li>Designed language composition from independently versioned packages: typed artifacts, capabilities and conflicts, deterministic routes, and pass ordering.</li><li>Implemented exact manifest binding, lock identity, fail-closed policies, and runtime-session lifecycle ownership.</li></ul></div>                                     | <div><div>Architecture</div><div>module boundaries, typed contracts, state machines, manifests/locks, lifecycle ownership, fail-closed design</div><div>.NET / backend</div><div>C#/.NET, ASP.NET Core, REST/OpenAPI, PostgreSQL, SQLite, webhooks, transactions, migrations, recovery</div><div>Compilers</div><div>Bytecode/AIR, CFG, SSA, dominance, optimizations, interpreter/CIL, LLVM, program analysis</div><div>Systems and verification</div><div>C++23, C17, Rust, Python, Linux, CMake, sanitizers, differential/metamorphic testing, exact oracles</div></div> |
| July 1 — August 31, 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <div><div>MCST — compiler engineering intern</div><ul style="list-style-type: none"><li>Developing an LLVM optimization pass and implementing tested graph algorithms in C++23.</li></ul></div>                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2023 — 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <div><div>Mama Lama — Software Developer</div><ul style="list-style-type: none"><li>Independently developed games and applications for DJs and events, from requirements and design through implementation, testing, and preparation for use.</li><li>Designed the application architecture and modular structure, defining component boundaries and interfaces; owned the complete technical implementation through a finished product.</li></ul></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| PROJECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EDUCATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <div><div>UniversalToolchain / Wist2</div><div>Semantic verification and backend-parity checks act as fail-closed regression gates, with verification records tied to exact revision/run identities.</div><div>VpnMediator</div><div>A production service with controlled state transitions, backup/restore, health gates, and safe rollback; the public case study excludes secrets and user data.</div><div>PS-form Memory Dependence Analyzer</div><div>Ranked 1st of 49 with the only 5.0/5.0 score and 104/104 official tests; the post-selection harness preserves reproducible wrong-answer/time-limit counterexamples.</div></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <div>Software Engineering student at HSE University</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                          | RECOGNITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <div>The Program Committee accepted “Build the Language, Then Make the Abstractions Disappear: Extensible Programming on .NET” for presentation on UniversalToolchain/Wist2 architecture in Torremolinos, Spain.</div> <div>Moscow / remote</div>                                                                                                                                                                                                                                                                                                                           |